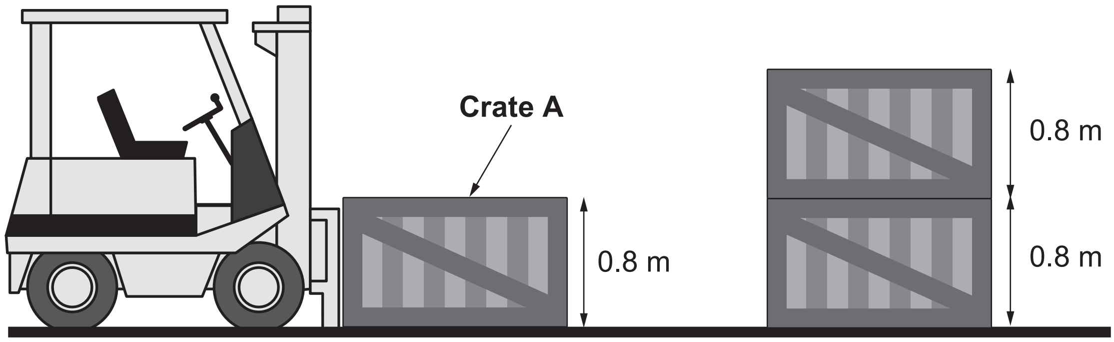
Examiner only

Answer **all** questions.

1. On a farm a fork-lift truck is used to stack wooden crates of Welsh apples.



(a) (i) Each crate of apples has a weight of 450 N and is 0.8 m high.
Use an equation from page 2 to calculate the work done in putting **crate A** on top of the **2 other** crates. [3]

Work done = ... J

(ii) State the gain in potential energy (PE) of the crate when lifted on to the stack. [1]

Gain in PE = ... J

(iii) Give a reason why the fork-lift truck uses more energy lifting crate A than the work done calculated in part (i). [1]

....

....

(b) Each crate of apples has a weight of 450 N.
The mass of the empty crate is 12 kg.
Calculate the mass of apples contained in the crate.
(On Earth, an object of weight 10 N has a mass of 1 kg.) [2]

Mass of apples = ... kg

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